

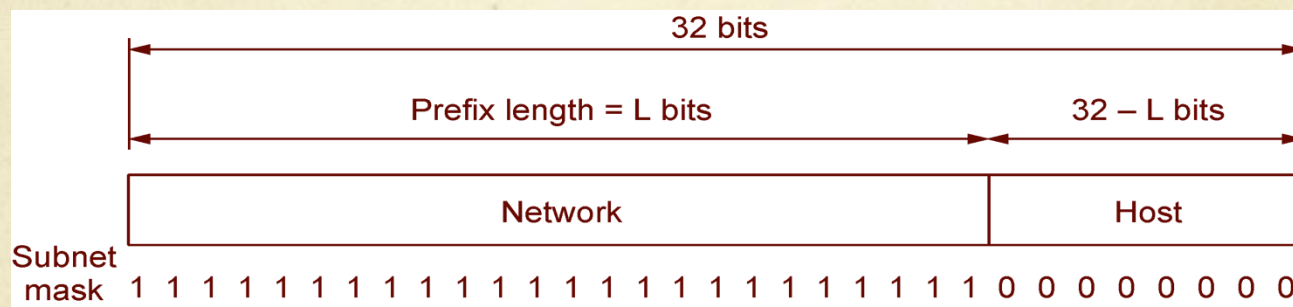
IP forwarding

- ◆ Packet must be forwarded based on *destination* IP address
- ◆ Should routing table have one entry *per* destination IP?
 - ◆ Routing table *explosion*!
- ◆ All IP addresses on one subnet have *same* prefix
 - ◆ Can combine destination addresses within subnet
 - ◆ Routing tables needs to store just *one* entry for each *prefix*

Prefix	Forward to
194.24.0.0/21	A
194.24.8.0/22	B
194.24.12.0/22	C
194.24.16.0/20	D

IP forwarding

- Router *matches* destination IP address w/ prefixes in table
 - IP address *and-ed* with *subnet mask* to extract prefix bits
 - Subnet mask contains 1 in all prefix bits & 0 for host bits



Prefix	Subnet mask	Forward to
194.24.0.0/21	255.255.248.0	A
194.24.8.0/22	255.255.252.0	B
194.24.12.0/22	255.255.252.0	C
194.24.16.0/20	255.255.240.0	D

IP forwarding

- ◆ One entry per subnet is also very large!
 - ◆ Route aggregation can be performed (*supernet*)

Location	Prefix	Subnet mask	Forward to
Cambridge	194.24.0.0/21	255.255.248.0	A
Edinburgh	194.24.8.0/22	255.255.252.0	B
San Francisco	194.24.12.0/22	255.255.252.0	C
Oxford	194.24.16.0/20	255.255.240.0	D

- ◆ Router in London keeps all entries
- ◆ Router in New York could *aggregate* entries 1, 2 and 4!

Location	Prefix	Subnet mask	Forward to
London	194.24.0.0/19	255.255.224.0	L
San Francisco	194.24.12.0/22	255.255.252.0	C

IP forwarding

Location	Prefix	Subnet mask	Forward to
London	194.24.0.0/19	255.255.224.0	L
San Francisco	194.24.12.0/22	255.255.252.0	C

- ◆ Destination IP address *194.24.16.150*
 - ◆ Forward to L
- ◆ Destination IP address *194.24.15.104*
 - ◆ Forward to C
- ◆ If prefixes overlap, forward based on *longest matching prefix*